var, x, y, z
$label$, L, $Next$, $Done$
lit
$blit$
f
$dsort$, D
$tvar$, a, b
$index$, i, j, k, n

terminals	::=		
	|	$\rightarrow$	
	|	$\rightsquigarrow$	
	|	$\rightarrow$	
	|	$\mid$	
	|	$\vee$	
$A,\ B,\ C$	::=		types
	|	int	integers
	|	bool	booleans
	|	$\mathsf{ptr}\ A$	pointer
	|	$(A_1,\ ..,A_n)$	record
	|	$D(A_1,\ ..,A_n)$	datatype application
	|	a	type variable
$\mathit{eff},\ \epsilon$	::=		effects
	|	$\mathbf{0}$	
	|	$\mathbf{1}$	
	|	$\mathbf{S}$	
τ	::=		sorts
	|	int	integers
	|	bool	booleans
	|	$\mathsf{ptr}\ \tau'$	pointer
	|	$(\tau_1,\ ..,\tau_n)$	tuple
	|	$D(\tau_1,\ ..,\tau_n)$	datasort application
	|	$\mathsf{pred}\ \tau'$	predicate sort
	|	a	type variable
F	::=		function signatures
	|	$\tau \rightarrow_{\mathit{eff}} \tau'$	
Γ	::=		contexts
	|	$\cdot$	
	|	$\Gamma, x{:}_{\mathit{eff}}\tau$	
	|	Γ, Γ'	
Σ_c	::=		core signatures
	|	$\cdot$	
	|	$\Sigma_c, f : F$	
	|	$\Sigma_c, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce$	
	|	$\Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}$	
	|	$\Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}$	
Σ_s	::=		surface signatures
	|	$\cdot$	
	|	$\Sigma_s, f : F$	
	|	$\Sigma_s, \mathsf{fun}\ f : \tau \rightarrow_{\mathit{eff}} \tau' = \{\overline{\mathbf{pat} \rightarrow \mathbf{se}}\}$	
	|	$\Sigma_s, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}$	

	$\mid$	$\Sigma_s, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}$	
Φ	$::=$		type variable contexts
	$\mid$	$\cdot$	
	$\mid$	Φ, a	
	$\mid$	$\overline{a}$	
$prim$	$::=$		primitives
	$\mid$	$\mathbf{Add}$	addition
	$\mid$	$\mathbf{Sub}$	subtraction
	$\mid$	$\mathbf{Mul}$	multiplication
	$\mid$	$\mathbf{Div}$	division
	$\mid$	$\mathbf{And}$	conjunction
	$\mid$	$\mathbf{Or}$	disjunction
	$\mid$	$\mathbf{Not}$	negation
	$\mid$	$\mathbf{Lt}$	less than
	$\mid$	$\mathbf{Le}$	less or equal
	$\mid$	$\mathbf{Gt}$	greater than
	$\mid$	$\mathbf{Ge}$	greater or equal
	$\mid$	$\mathbf{Eq}\,[A]$	equality test
	$\mid$	$\mathbf{New}\,[A]$	allocate
	$\mid$	$\mathbf{Del}\,[A]$	deallocate
	$\mid$	$\mathbf{Get}\,[A]$	dereference
	$\mid$	$\mathbf{Set}\,[A]$	update
	$\mid$	$\mathbf{Own}\,[A]$	ownership
u	$::=$		usage flags
	$\mid$	$\mathbf{0}$	
	$\mid$	$\mathbf{1}$	
Δ	$::=$		refined contexts
	$\mid$	$\cdot$	
	$\mid$	$\Delta, x{:}_{\mathit{eff}}\tau$	
	$\mid$	Δ, Δ'	
	$\mid$	$\Delta, \mathsf{log}\ x{:}ce$	
	$\mid$	$\Delta, \mathsf{res}\ x{:}^{u}ce@ce'$	
X	$::=$		lookup results
	$\mid$	τ	
	$\mid$	ce	
	$\mid$	$ce@ce'$	
Π	$::=$		preconditions
	$\mid$	$\cdot$	
	$\mid$	$\mathbf{take}\, x = ce; \Pi$	
	$\mid$	$\mathbf{let}\, x = ce; \Pi$	
	$\mid$	$\mathbf{let}\, (x_1, \ldots, x_n) = ce; \Pi$	
$\mathcal{F}$	$::=$		refined type schemes

	$\mid$	$\forall a{:}\tau.\ \Pi x{:}A.\ \Pi \to \exists b{:}\tau'.\ \Sigma y{:}B.\ z{:}ce\ \mathbf{eff}]$	
ce	$::=$		core expressions
	$\mid$	x	
	$\mid$	lit	integer literal
	$\mid$	$blit$	boolean literal
	$\mid$	$\mathbf{let}\, x = ce_1; ce_2$	
	$\mid$	$(ce_1, .., ce_n)$	
	$\mid$	$\mathbf{let}\, (x_1, .., x_n) = ce_1; ce_2$	
	$\mid$	$L\ ce$	
	$\mid$	$\mathbf{case}\, ce\, \mathbf{of}\, \{L_1\, x_1 \to ce_1 \mid .. \mid L_n\, x_n \to ce_n\}$	
	$\mid$	$\mathbf{iter}\, (x = ce_1)\{ce_2\}$	
	$\mid$	$prim\ ce$	
	$\mid$	$\mathbf{if}\, ce_1\, \mathbf{then}\, ce_2\, \mathbf{else}\, ce_3$	conditional
	$\mid$	$ce : \tau$	
	$\mid$	$f\ ce$	
	$\mid$	$ce_1 \doteq ce_2$	
	$\mid$	$\mathbf{take}\, x = ce_1; ce_2$	
	$\mid$	$\mathbf{return}\, ce$	
se	$::=$		surface expressions
	$\mid$	x	
	$\mid$	lit	integer literal
	$\mid$	$blit$	boolean literal
	$\mid$	$\mathbf{let}\, pat = se_1; se_2$	
	$\mid$	$(se_1, .., se_n)$	
	$\mid$	$L\ se$	
	$\mid$	$\mathbf{case}\, se\, \mathbf{of}\, \{pat_1 \to se_1 \mid .. \mid pat_n \to se_n\}$	
	$\mid$	$\mathbf{iter}\, (pat = se_1)\{se_2\}$	
	$\mid$	$prim\ se$	
	$\mid$	$\mathbf{if}\, se_1\, \mathbf{then}\, se_2\, \mathbf{else}\, se_3$	conditional
	$\mid$	$se : \tau$	
	$\mid$	$f\ se$	
	$\mid$	$se_1 \doteq se_2$	
	$\mid$	$\mathbf{take}\, pat = se_1; se_2$	
	$\mid$	$\mathbf{return}\, se$	
$prog$	$::=$		surface programs
	$\mid$	$\mathsf{fun}\, f : \tau \to_{eff} \tau' = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \quad prog$	
	$\mid$	$\mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\} prog$	
	$\mid$	$\mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 \mid .. \mid L_n : A_n\} prog$	
	$\mid$	$\mathsf{main} : \tau_{eff} = se$	
$cprog$	$::=$		core programs
	$\mid$	$\mathsf{fun}\, f(x : \tau) \to_{eff} \tau' = ce \quad cprog$	
	$\mid$	$\mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\} cprog$	
	$\mid$	$\mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 \mid .. \mid L_n : A_n\} cprog$	
	$\mid$	$\mathsf{main} : \tau_{eff} = ce$	
pat	$::=$		patterns

	$\mid$	x	
	$\mid$	$L\,pat$	
	$\mid$	$(pat_1, \ldots, pat_n)$	
β	$::=$		pattern bindings
	$\mid$	$\cdot$	
	$\mid$	$pat : \tau, \beta$	
	$\mid$	$pat_1 : \tau_1, \ldots, pat_n : \tau_n, \beta$	
	$\mid$	$pat : \tau$	
$\mathcal{E}$	$::=$		evaluation contexts
	$\mid$	$[\cdot]$	
	$\mid$	$\mathbf{let}\, x = ce; \mathcal{E}$	
$\overline{y}$	$::=$		scrutinee variables
	$\mid$	$\cdot$	
	$\mid$	$y, \overline{y}$	
	$\mid$	$y_1, \ldots, y_n, \overline{y}$	
M	$::=$		match branches
	$\mid$	$\cdot$	
	$\mid$	$\beta \rightarrow [\Gamma][\mathcal{E}]se \mid M$	
	$\mid$	$\beta \rightarrow [\Gamma][\mathcal{E}]se$	
$formula$	$::=$		
	$\mid$	$judgement$	
	$\mid$	$x{:}_{eff}\tau \in \Gamma$	
	$\mid$	$f{:}F \in \Sigma_c$	
	$\mid$	$A\ \mathsf{eqtype}$	
	$\mid$	$\mathbf{sort}\, D(a_1, \ldots, a_i)\, \mathbf{in}\, \Sigma_c$	
	$\mid$	$L{:}\tau \in \mathrm{ctors}(D) \in \Sigma_c$	
	$\mid$	$\tau = \mathbf{subst}\, \tau_1/a_1, \ldots, \tau_i/a_i\, \mathbf{in}\, \tau'$	
	$\mid$	$\mathbf{sort}\, D(a_1, \ldots, a_k) = \{L_1 : \tau_1 \mid \ldots \mid L_n : \tau_n\}\, \mathbf{in}\, \Sigma_c$	
	$\mid$	$\mathbf{type}\, D(a_1, \ldots, a_i)\, \mathbf{in}\, \Sigma_c$	
	$\mid$	$L{:}A \in \mathsf{tyctors}(D) \in \Sigma_c$	
	$\mid$	$A = \mathbf{subst}\, A_1/a_1, \ldots, A_i/a_i\, \mathbf{in}\, A'$	
	$\mid$	$\mathbf{type}\, D(a_1, \ldots, a_k) = \{L_1 : A_1 \mid \ldots \mid L_n : A_n\}\, \mathbf{in}\, \Sigma_c$	
	$\mid$	$\mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c$	
	$\mid$	$a \in \Phi$	
	$\mid$	$L \neq L'$	
	$\mid$	$M = \overline{pat{:}\tau, \cdot \rightarrow [\cdot][\cdot]\, se}$	
	$\mid$	$ce = \mathcal{E}[ce']$	
	$\mid$	$D \neq D'$	
	$\mid$	$x \neq y$	
	$\mid$	$n = k$	
	$\mid$	$eff = eff_1 \vee eff_2$	
	$\mid$	$eff = eff_1 \vee eff_2 \vee eff_3$	
	$\mid$	$eff = eff' \vee eff_1 \vee \ldots \vee eff_n$	
	$\mid$	$eff = eff_1 \vee \ldots \vee eff_n$	
	$\mid$	$formula_1 \quad \ldots \quad formula_n$	

Jtype	::=	
	|	$\Sigma_c; \Phi \vdash A$ wf
	|	$\Sigma_c; \Phi; D' \vdash A$ guarded
	|	$\Sigma_c \vdash$ sort $D(\overline{a}) = \{\overline{L : \tau}\}$ ok
	|	$\Sigma_c \vdash$ type $D(\overline{a}) = \{\overline{L : A}\}$ ok
	|	$\Sigma_c; \Phi \vdash \tau$ wf
	|	A eqtype
	|	$\mathit{eff} \leq \mathit{eff}'$
	|	$\mathit{eff}' = \lfloor \mathit{eff} \rfloor$
	|	$\mathit{eff} = \mathit{eff}_1 \vee \mathit{eff}_2$
	|	$\mathit{prim}{:}\tau \rightarrow_{\mathit{eff}} \tau'$
	|	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} \Rightarrow \tau\ [\mathit{eff}]$
	|	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} \Leftarrow \tau\ [\mathit{eff}]$
	|	$\Sigma_c \vdash \mathit{cprog}$
Jcore	::=	
	|	$L : \tau\ \mathbf{in}\ D(\tau_1, .., \tau_i)\ \mathbf{in}\ \Sigma_c$
	|	$D(\overline{a}) = \{\overline{L : \tau}\} \in ..$
Jelaborate	::=	
	|	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se} \Rightarrow \tau\ [\mathit{eff}] \rightsquigarrow \mathit{ce}$
	|	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se} \Leftarrow \tau\ [\mathit{eff}] \rightsquigarrow \mathit{ce}$
	|	$\Sigma_s \vdash \mathit{prog}$ $\rightsquigarrow \Sigma_c \vdash \mathit{cprog}$
Jmatch	::=	
	|	M has_con
	|	M has_tup
	|	$M/\mathit{var} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'$
	|	$M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'$
	|	$M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'$
	|	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \overline{y}; M \Leftarrow_\epsilon \tau\ [\mathit{eff}] \rightsquigarrow \mathit{ce}$
	|	$\mathcal{E}[\mathit{ce}] \rightsquigarrow \mathit{ce}'$
Jrlookup	::=	
	|	$\Delta \vdash x \Rightarrow X \dashv \Delta'$
	|	$\Sigma_c \vdash \Delta$ wf
judgement	::=	
	|	*Jtype*
	|	*Jcore*
	|	*Jelaborate*
	|	*Jmatch*
	|	*Jrlookup*
user_syntax	::=	
	|	*var*
	|	*label*
	|	*lit*
	|	*blit*

$$
\begin{array}{|l}
f \\
dsort \\
tvar \\
index \\
terminals \\
A \\
eff \\
\tau \\
F \\
\Gamma \\
\Sigma_c \\
\Sigma_s \\
\Phi \\
prim \\
u \\
\Delta \\
X \\
\Pi \\
\mathcal{F} \\
ce \\
se \\
prog \\
cprog \\
pat \\
\beta \\
\mathcal{E} \\
\overline{y} \\
M \\
formula
\end{array}
$$

$\boxed{\Sigma_c; \Phi \vdash A \ \mathsf{wf}}$

$$\frac{}{\Sigma_c; \Phi \vdash \mathsf{int\ wf}} \quad \textsc{TWf_int}$$

$$\frac{}{\Sigma_c; \Phi \vdash \mathsf{bool\ wf}} \quad \textsc{TWf_bool}$$

$$\frac{\Sigma_c; \Phi \vdash A \ \mathsf{wf}}{\Sigma_c; \Phi \vdash \mathsf{ptr}\ A \ \mathsf{wf}} \quad \textsc{TWf_ptr}$$

$$\frac{\Sigma_c; \Phi \vdash A_1 \ \mathsf{wf} \quad .. \quad \Sigma_c; \Phi \vdash A_n \ \mathsf{wf}}{\Sigma_c; \Phi \vdash (A_1, \, .., \, A_n) \ \mathsf{wf}} \quad \textsc{TWf_record}$$

$$\frac{\begin{array}{c}\mathbf{type}\, D(a_1, \, .., \, a_k)\, \mathbf{in}\, \Sigma_c \quad n = k \\ \Sigma_c; \Phi \vdash A_1 \ \mathsf{wf} \quad .. \quad \Sigma_c; \Phi \vdash A_n \ \mathsf{wf}\end{array}}{\Sigma_c; \Phi \vdash D(A_1, \, .., \, A_n) \ \mathsf{wf}} \quad \textsc{TWf_app}$$

$$\frac{a \in \Phi}{\Sigma_c; \Phi \vdash a \ \mathsf{wf}} \quad \textsc{TWf_tvar}$$

$\boxed{\Sigma_c; \Phi; D' \vdash A \ \mathsf{guarded}}$

$$\frac{}{\Sigma_c; \Phi; D' \vdash \mathsf{int\ guarded}} \quad \textsc{TWfG_int}$$

$$\frac{}{\Sigma_c; \Phi; D' \vdash \mathsf{bool\ guarded}} \quad \textsc{TWfG_bool}$$

$$\frac{\Sigma_c; \Phi \vdash A\ \mathsf{wf}}{\Sigma_c; \Phi; D' \vdash \mathsf{ptr}\ A\ \mathsf{guarded}} \quad \textsc{TWfG_ptr}$$

$$\frac{\Sigma_c; \Phi; D' \vdash A_1\ \mathsf{guarded} \quad .. \quad \Sigma_c; \Phi; D' \vdash A_n\ \mathsf{guarded}}{\Sigma_c; \Phi; D' \vdash (A_1, .., A_n)\ \mathsf{guarded}} \quad \textsc{TWfG_record}$$

$$\frac{\begin{array}{c}\textbf{type}\, D(a_1, .., a_k)\, \textbf{in}\, \Sigma_c \quad n = k \quad D \neq D' \\ \Sigma_c; \Phi \vdash A_1\ \mathsf{wf} \quad .. \quad \Sigma_c; \Phi \vdash A_n\ \mathsf{wf}\end{array}}{\Sigma_c; \Phi; D' \vdash D(A_1, .., A_n)\ \mathsf{guarded}} \quad \textsc{TWfG_app}$$

$$\frac{a \in \Phi}{\Sigma_c; \Phi; D' \vdash a\ \mathsf{guarded}} \quad \textsc{TWfG_tvar}$$

$\boxed{\Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok}}$

$$\frac{\Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}; \overline{a} \vdash \tau_1\ \mathsf{wf} \quad .. \quad \Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}; \overline{a} \vdash \tau_n\ \mathsf{wf}}{\Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok}} \quad \textsc{SDWf_ok}$$

$\boxed{\Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok}}$

$$\frac{\Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}; \overline{a}; D \vdash A_1\ \mathsf{guarded} \quad .. \quad \Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}; \overline{a}; D \vdash A_n\ \mathsf{guarded}}{\Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok}} \quad \textsc{TDWf_ok}$$

$\boxed{\Sigma_c; \Phi \vdash \tau\ \mathsf{wf}}$

$$\frac{}{\Sigma_c; \Phi \vdash \mathsf{int\ wf}} \quad \textsc{SWf_int}$$

$$\frac{}{\Sigma_c; \Phi \vdash \mathsf{bool\ wf}} \quad \textsc{SWf_bool}$$

$$\frac{\Sigma_c; \Phi \vdash \tau\ \mathsf{wf}}{\Sigma_c; \Phi \vdash \mathsf{ptr}\ \tau\ \mathsf{wf}} \quad \textsc{SWf_ptr}$$

$$\frac{\Sigma_c; \Phi \vdash \tau_1\ \mathsf{wf} \quad .. \quad \Sigma_c; \Phi \vdash \tau_n\ \mathsf{wf}}{\Sigma_c; \Phi \vdash (\tau_1, .., \tau_n)\ \mathsf{wf}} \quad \textsc{SWf_record}$$

$$\frac{\begin{array}{c}\textbf{sort}\, D(a_1, .., a_k)\, \textbf{in}\, \Sigma_c \quad n = k \\ \Sigma_c; \Phi \vdash \tau_1\ \mathsf{wf} \quad .. \quad \Sigma_c; \Phi \vdash \tau_n\ \mathsf{wf}\end{array}}{\Sigma_c; \Phi \vdash D(\tau_1, .., \tau_n)\ \mathsf{wf}} \quad \textsc{SWf_app}$$

$$\frac{\Sigma_c; \Phi \vdash \tau\ \mathsf{wf}}{\Sigma_c; \Phi \vdash \mathsf{pred}\ \tau\ \mathsf{wf}} \quad \textsc{SWf_pred}$$

$$\frac{a \in \Phi}{\Sigma_c; \Phi \vdash a\ \mathsf{wf}} \quad \textsc{SWf_tvar}$$

$\boxed{A\ \mathsf{eqtype}}$

$$\frac{}{\mathsf{int\ eqtype}} \quad \textsc{EqTy_int}$$

$$\frac{}{\mathsf{bool\ eqtype}} \quad \textsc{EqTy_bool}$$

$$\frac{}{\mathsf{ptr}\ A\ \mathsf{eqtype}} \quad \textsc{EqTy_ptr}$$

$\boxed{\mathit{eff} \leq \mathit{eff}'}$

$$\frac{}{\mathbf{0} \leq \mathbf{0}} \quad \textsc{Sub_pure_pure}$$

$$\frac{}{\mathbf{0} \leq \mathbf{1}} \quad \textsc{Sub_pure_impure}$$

$$\frac{}{\mathbf{1} \leq \mathbf{1}} \quad \textsc{Sub_impure_impure}$$

$$\frac{}{\mathbf{0} \leq \mathbf{S}} \quad \textsc{Sub_pure_spec}$$

$$\frac{}{\mathbf{S} \leq \mathbf{S}} \quad \textsc{Sub_spec_spec}$$

$\boxed{\mathit{eff}' = \lfloor \mathit{eff} \rfloor}$

$$\frac{}{\mathbf{0} = \lfloor \mathbf{0} \rfloor} \quad \textsc{Pur_pure}$$

$$\frac{}{\mathbf{0} = \lfloor \mathbf{1} \rfloor} \quad \textsc{Pur_impure}$$

$$\frac{}{\mathbf{S} = \lfloor \mathbf{S} \rfloor} \quad \textsc{Pur_spec}$$

$\boxed{\mathit{eff} = \mathit{eff}_1 \vee \mathit{eff}_2}$

$$\frac{}{\mathit{eff} = \mathit{eff} \vee \mathit{eff}} \quad \textsc{Join_refl}$$

$$\frac{}{\mathbf{1} = \mathbf{0} \vee \mathbf{1}} \quad \textsc{Join_pure_impure}$$

$$\frac{}{\mathbf{1} = \mathbf{1} \vee \mathbf{0}} \quad \textsc{Join_impure_pure}$$

$$\frac{}{\mathbf{S} = \mathbf{0} \vee \mathbf{S}} \quad \textsc{Join_pure_spec}$$

$$\frac{}{\mathbf{S} = \mathbf{S} \vee \mathbf{0}} \quad \textsc{Join_spec_pure}$$

$\boxed{\mathit{prim}{:}\tau \rightarrow_{\mathit{eff}} \tau'}$

$$\frac{}{\mathbf{Add}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{int}} \quad \textsc{PT_add}$$

$$\frac{}{\mathbf{Sub}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{int}} \quad \textsc{PT_sub}$$

$$\frac{}{\mathbf{Mul}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{int}} \quad \textsc{PT_mul}$$

$$\frac{}{\mathbf{Div}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{1}} \mathsf{int}} \quad \textsc{PT_div}$$

$$\frac{}{\mathbf{And}{:}(\mathsf{bool}, \mathsf{bool}) \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \textsc{PT_and}$$

$$\frac{}{\mathbf{Or}{:}(\mathsf{bool}, \mathsf{bool}) \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \textsc{PT_or}$$

$$\frac{}{\mathbf{Not}{:}\mathsf{bool} \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \textsc{PT_not}$$

$$\frac{}{\mathbf{Lt}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \textsc{PT_lt}$$

$$\frac{}{\mathbf{Le}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \textsc{PT_le}$$

$$\frac{}{\mathbf{Gt}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \textsc{PT_gt}$$

$$\frac{}{\mathbf{Ge}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \text{PT_GE}$$

$$\frac{A \ \mathsf{eqtype}}{\mathbf{Eq}\,[A]{:}(A, A) \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \text{PT_EQ}$$

$$\frac{}{\mathbf{New}\,[A]{:}A \rightarrow_{\mathbf{1}} \mathsf{ptr}\ A} \quad \text{PT_NEW}$$

$$\frac{}{\mathbf{Del}\,[A]{:}\mathsf{ptr}\ A \rightarrow_{\mathbf{1}} (\,)} \quad \text{PT_DEL}$$

$$\frac{}{\mathbf{Get}\,[A]{:}\mathsf{ptr}\ A \rightarrow_{\mathbf{1}} A} \quad \text{PT_GET}$$

$$\frac{}{\mathbf{Set}\,[A]{:}(\mathsf{ptr}\ A, A) \rightarrow_{\mathbf{1}} (\,)} \quad \text{PT_SET}$$

$$\frac{}{\mathbf{Own}\,[A]{:}\mathsf{ptr}\ A \rightarrow_{\mathbf{S}} \mathsf{pred}\ (A)} \quad \text{PT_OWN}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} \Rightarrow \tau\ [\mathit{eff}]}$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{lit} \Rightarrow \mathsf{int}\ [\mathbf{0}]} \quad \text{SYN_INT}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{blit} \Rightarrow \mathsf{bool}\ [\mathbf{0}]} \quad \text{SYN_BOOL}$$

$$\frac{x{:}_{\mathit{eff}}\tau \in \Gamma \quad \mathit{eff} \leq \mathit{eff}_0}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x \Rightarrow \tau\ [\mathit{eff}]} \quad \text{SYN_VAR}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} \Leftarrow \tau\ [\mathit{eff}]}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} : \tau \Rightarrow \tau\ [\mathit{eff}]} \quad \text{SYN_ANNOT}$$

$$\frac{\begin{array}{l} \mathit{prim}{:}\tau \rightarrow_{\mathit{eff}'} \tau' \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} \mathit{ce} \Leftarrow \tau\ [\mathit{eff}_1] \\ \mathit{eff} = \mathit{eff}' \vee \mathit{eff}_1 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{prim}\ \mathit{ce} \Rightarrow \tau'\ [\mathit{eff}]} \quad \text{SYN_APP}$$

$$\frac{\begin{array}{l} f{:}\tau \rightarrow_{\mathit{eff}'} \tau' \in \Sigma_c \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} \mathit{ce} \Leftarrow \tau\ [\mathit{eff}_1] \\ \mathit{eff} = \mathit{eff}' \vee \mathit{eff}_1 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} f\ \mathit{ce} \Rightarrow \tau'\ [\mathit{eff}]} \quad \text{SYN_CALL}$$

$$\frac{\begin{array}{l} A \ \mathsf{eqtype} \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce}_1 \Rightarrow A\ [\mathit{eff}'] \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce}_2 \Leftarrow A\ [\mathit{eff}''] \\ \mathit{eff} = \mathit{eff}' \vee \mathit{eff}'' \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce}_1 \doteq \mathit{ce}_2 \Rightarrow \mathsf{bool}\ [\mathit{eff}]} \quad \text{SYN_EQ}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} \Leftarrow \tau\ [\mathit{eff}]}$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce}_1 \Rightarrow \tau\ [\mathit{eff}'] \\ \epsilon = \lfloor \mathit{eff}' \rfloor \\ \Sigma_c; \Gamma, x{:}_{\epsilon}\tau \vdash_{\mathit{eff}_0} \mathit{ce}_2 \Leftarrow \tau'\ [\mathit{eff}''] \\ \mathit{eff} = \mathit{eff}' \vee \mathit{eff}'' \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{let}\ x = \mathit{ce}_1; \mathit{ce}_2 \Leftarrow \tau'\ [\mathit{eff}]} \quad \text{CHK_LET}$$

$$\frac{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce_1 \Leftarrow \tau_1\ [\mathit{eff}_1] \quad .. \quad \Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce_n \Leftarrow \tau_n\ [\mathit{eff}_n] \quad \mathit{eff} = \mathit{eff}_1 \lor .. \lor \mathit{eff}_n}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} (ce_1, .., ce_n) \Leftarrow (\tau_1, .., \tau_n)\ [\mathit{eff}]} \quad \textsc{Chk_tuple}$$

$$\frac{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce_1 \Rightarrow (\tau_1, .., \tau_n)\ [\mathit{eff}'] \quad \epsilon = \lfloor \mathit{eff}' \rfloor \quad \Sigma_c;\Gamma' \vdash_{\mathit{eff}_0} ce_2 \Leftarrow \tau'\ [\mathit{eff}''] \quad \mathit{eff} = \mathit{eff}' \lor \mathit{eff}''}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} \mathbf{let}\,(x_1, .., x_n) = ce_1; ce_2 \Leftarrow \tau'\ [\mathit{eff}]} \quad \textsc{Chk_lettuple}$$

$$\frac{L : \tau\,\mathbf{in}\,D(\tau_1, .., \tau_i)\,\mathbf{in}\,\Sigma_c \quad \Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce \Leftarrow \tau\ [\mathit{eff}]}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} L\,ce \Leftarrow D(\tau_1, .., \tau_i)\ [\mathit{eff}]} \quad \textsc{Chk_inject}$$

$$\frac{\begin{array}{l} \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c;\Gamma \vdash_{\mathit{eff}'} ce \Rightarrow D(\tau_1, .., \tau_i)\ [\mathit{eff}''] \\ D(\overline{a}) = \{\overline{L : \tau}\} \in .. \\ L_1 : \tau_1''\,\mathbf{in}\,D(\tau_1, .., \tau_i)\,\mathbf{in}\,\Sigma_c \quad .. \quad L_n : \tau_n''\,\mathbf{in}\,D(\tau_1, .., \tau_i)\,\mathbf{in}\,\Sigma_c \\ \Sigma_c;\Gamma, x_1{:}_{\mathit{eff}''}\tau_1'' \vdash_{\mathit{eff}_0} ce_1 \Leftarrow \tau\ [\mathit{eff}_1] \quad .. \quad \Sigma_c;\Gamma, x_n{:}_{\mathit{eff}''}\tau_n'' \vdash_{\mathit{eff}_0} ce_n \Leftarrow \tau\ [\mathit{eff}_n] \\ \mathit{eff} = \mathit{eff}'' \lor \mathit{eff}_1 \lor .. \lor \mathit{eff}_n \end{array}}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} \mathbf{case}\,ce\,\mathbf{of}\,\{L_1\,x_1 \to ce_1 | .. | L_n\,x_n \to ce_n\} \Leftarrow \tau\ [\mathit{eff}]} \quad \textsc{Chk_case}$$

$$\frac{\begin{array}{l} \mathbf{1} \leq \mathit{eff}_0 \\ \mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c \\ \Sigma_c;\Gamma \vdash_{\mathbf{0}} ce_1 \Rightarrow A\ [\mathbf{0}] \\ \Sigma_c;\Gamma, x{:}_{\mathbf{0}}A \vdash_{\mathbf{1}} ce_2 \Leftarrow D(A, B)\ [\mathit{eff}'] \end{array}}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} \mathbf{iter}\,(x = ce_1)\{ce_2\} \Leftarrow B\ [\mathbf{1}]} \quad \textsc{Chk_iter}$$

$$\frac{\begin{array}{l} \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c;\Gamma \vdash_{\mathit{eff}'} ce_1 \Leftarrow \mathsf{bool}\ [\mathit{eff}_1] \\ \Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce_2 \Leftarrow \tau\ [\mathit{eff}_2] \\ \Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce_3 \Leftarrow \tau\ [\mathit{eff}_3] \\ \mathit{eff} = \mathit{eff}_1 \lor \mathit{eff}_2 \lor \mathit{eff}_3 \end{array}}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} \mathbf{if}\,ce_1\,\mathbf{then}\,ce_2\,\mathbf{else}\,ce_3 \Leftarrow \tau\ [\mathit{eff}]} \quad \textsc{Chk_if}$$

$$\frac{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce \Leftarrow \tau\ [\mathit{eff}]}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} \mathbf{return}\,ce \Leftarrow \mathsf{pred}\ \tau\ [\mathit{eff}]} \quad \textsc{Chk_return}$$

$$\frac{\begin{array}{l} \Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce_1 \Rightarrow \mathsf{pred}\ \tau\ [\mathit{eff}'] \\ \epsilon = \lfloor \mathit{eff}' \rfloor \\ \Sigma_c;\Gamma, x{:}_{\epsilon}\tau \vdash_{\mathit{eff}_0} ce_2 \Leftarrow \mathsf{pred}\ \tau'\ [\mathit{eff}''] \\ \mathit{eff} = \mathit{eff}' \lor \mathit{eff}'' \end{array}}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} \mathbf{take}\,x = ce_1; ce_2 \Leftarrow \mathsf{pred}\ \tau'\ [\mathit{eff}]} \quad \textsc{Chk_take}$$

$$\frac{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce \Rightarrow \tau\ [\mathit{eff}]}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce \Leftarrow \tau\ [\mathit{eff}]} \quad \textsc{Chk_sub}$$

$\boxed{\Sigma_c \vdash \mathit{cprog}}$

$$\frac{\Sigma_c;\Gamma \vdash_{\mathit{eff}} ce \Leftarrow \tau\ [\mathit{eff}']}{\Sigma_c \vdash \mathsf{main} : \tau_{\mathit{eff}} = ce} \quad \textsc{CProg_main}$$

$$\frac{\begin{array}{l} \Sigma_c;\Gamma, x{:}_{\mathbf{0}}A \vdash_{\mathbf{0}} ce \Leftarrow B\ [\mathit{eff}'] \\ \Sigma_c, \mathsf{fun}\ f(x : A) \to_{\mathbf{0}} B = ce \vdash \mathit{cprog} \end{array}}{\Sigma_c \vdash \mathsf{fun}\ f(x : A) \to_{\mathbf{0}} B = ce \quad \mathit{cprog}} \quad \textsc{CProg_purefun}$$

$$\frac{\Sigma_c, f : A \to_{\mathbf{1}} B; \Gamma, x{:}_{\mathbf{0}} A \vdash_{\mathbf{1}} ce \Leftarrow B\ [\mathit{eff}'] \quad \Sigma_c, f : A \to_{\mathbf{1}} B \vdash \mathit{cprog}}{\Sigma_c \vdash \mathsf{fun}\ f(x : A) \to_{\mathbf{1}} B = ce \quad \mathit{cprog}} \quad \textsc{CProg_impurefun}$$

$$\frac{\Sigma_c, f : \tau \to_{\mathbf{S}} \tau'; \Gamma, x{:}_{\mathbf{S}} \tau \vdash_{\mathbf{S}} ce \Leftarrow \tau'\ [\mathit{eff}'] \quad \Sigma_c, \mathsf{fun}\ f(x : \tau) \to_{\mathbf{S}} \tau' = ce \vdash \mathit{cprog}}{\Sigma_c \vdash \mathsf{fun}\ f(x : \tau) \to_{\mathbf{S}} \tau' = ce \quad \mathit{cprog}} \quad \textsc{CProg_specfun}$$

$$\frac{\Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok} \quad \Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} \vdash \mathit{cprog}}{\Sigma_c \vdash \mathbf{sort}\ D(a_1, .., a_i) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} \mathit{cprog}} \quad \textsc{CProg_sortdecl}$$

$$\frac{\Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok} \quad \Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} \vdash \mathit{cprog}}{\Sigma_c \vdash \mathbf{type}\ D(a_1, .., a_i) = \{L_1 : A_1 | .. | L_n : A_n\} \mathit{cprog}} \quad \textsc{CProg_typedecl}$$

$\boxed{L : \tau\ \mathbf{in}\ D(\tau_1, .., \tau_i)\ \mathbf{in}\ \Sigma_c}$

$$\frac{\mathbf{sort}\ D(a_1, .., a_i)\ \mathbf{in}\ \Sigma_c \quad L{:}\tau' \in \mathrm{ctors}(D) \in \Sigma_c \quad \tau = \mathbf{subst}\ \tau_1/a_1, .., \tau_i/a_i\ \mathbf{in}\ \tau'}{L : \tau\ \mathbf{in}\ D(\tau_1, .., \tau_i)\ \mathbf{in}\ \Sigma_c} \quad \textsc{Ctor_sort}$$

$$\frac{\mathbf{type}\ D(a_1, .., a_i)\ \mathbf{in}\ \Sigma_c \quad L{:}A' \in \mathsf{tyctors}(D) \in \Sigma_c \quad A = \mathbf{subst}\ A_1/a_1, .., A_i/a_i\ \mathbf{in}\ A'}{L : A\ \mathbf{in}\ D(A_1, .., A_i)\ \mathbf{in}\ \Sigma_c} \quad \textsc{Ctor_type}$$

$\boxed{D(\overline{a}) = \{\overline{L : \tau}\} \in ..}$

$$\frac{\mathbf{sort}\ D(a_1, .., a_k) = \{L_1 : \tau_1 | .. | L_n : \tau_n\}\ \mathbf{in}\ \Sigma_c}{D(\overline{a}) = \{\overline{L : \tau}\} \in ..} \quad \textsc{DL_sort}$$

$$\frac{\mathbf{type}\ D(a_1, .., a_k) = \{L_1 : A_1 | .. | L_n : A_n\}\ \mathbf{in}\ \Sigma_c}{D(\overline{a}) = \{\overline{L : \tau}\} \in ..} \quad \textsc{DL_type}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Rightarrow \tau\ [\mathit{eff}] \rightsquigarrow ce}$

$$\frac{x{:}_{\mathit{eff}}\tau \in \Gamma \quad \mathit{eff} \leq \mathit{eff}_0}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x \Rightarrow \tau\ [\mathit{eff}] \rightsquigarrow x} \quad \textsc{SurfSyn_var}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{lit} \Rightarrow \mathsf{int}\ [\mathbf{0}] \rightsquigarrow \mathit{lit}} \quad \textsc{SurfSyn_int}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{blit} \Rightarrow \mathsf{bool}\ [\mathbf{0}] \rightsquigarrow \mathit{blit}} \quad \textsc{SurfSyn_bool}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau\ [\mathit{eff}] \rightsquigarrow ce}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se : \tau \Rightarrow \tau\ [\mathit{eff}] \rightsquigarrow ce : \tau} \quad \textsc{SurfSyn_annot}$$

$$\frac{\begin{array}{l} \mathit{prim}{:}\tau \to_{\mathit{eff}'} \tau' \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} se \Leftarrow \tau\ [\mathit{eff}_1] \rightsquigarrow ce \\ \mathit{eff} = \mathit{eff}' \vee \mathit{eff}_1 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{prim}\ se \Rightarrow \tau'\ [\mathit{eff}] \rightsquigarrow \mathit{prim}\ ce} \quad \textsc{SurfSyn_app}$$

$$\frac{\begin{array}{l} f{:}\tau \to_{\mathit{eff}'} \tau' \in \Sigma_c \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} se \Leftarrow \tau\ [\mathit{eff}_1] \rightsquigarrow ce \\ \mathit{eff} = \mathit{eff}' \vee \mathit{eff}_1 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} f\ se \Rightarrow \tau'\ [\mathit{eff}] \rightsquigarrow f\ ce} \quad \textsc{SurfSyn_call}$$

$$\frac{\begin{array}{l} A \ \mathsf{eqtype} \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_1 \Rightarrow A\ [\mathit{eff}'] \rightsquigarrow ce_1 \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_2 \Leftarrow A\ [\mathit{eff}''] \rightsquigarrow ce_2 \\ \mathit{eff} = \mathit{eff}' \vee \mathit{eff}'' \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_1 \doteq se_2 \Rightarrow \mathsf{bool}\ [\mathit{eff}] \rightsquigarrow ce_1 \doteq ce_2} \quad \textsc{SurfSyn_eq}$$

$$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau\ [\mathit{eff}] \rightsquigarrow ce}$$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_1 \Rightarrow \tau\ [\mathit{eff}'] \rightsquigarrow ce_1 \\ \epsilon = \lfloor \mathit{eff}' \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \cdot; pat : \tau \to [\cdot][[\cdot]] se_2 \Leftarrow_\epsilon \tau'\ [\mathit{eff}''] \rightsquigarrow ce_2 \\ \mathit{eff} = \mathit{eff}' \vee \mathit{eff}'' \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{let}\, pat = se_1; se_2 \Leftarrow \tau'\ [\mathit{eff}] \rightsquigarrow \mathbf{let}\, y = ce_1; ce_2} \quad \textsc{SurfChk_let}$$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_1 \Leftarrow \tau_1\ [\mathit{eff}_1] \rightsquigarrow ce_1 \quad .. \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_n \Leftarrow \tau_n\ [\mathit{eff}_n] \rightsquigarrow ce_n \\ \mathit{eff} = \mathit{eff}_1 \vee .. \vee \mathit{eff}_n \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} (se_1, .., se_n) \Leftarrow (\tau_1, .., \tau_n)\ [\mathit{eff}] \rightsquigarrow (ce_1, .., ce_n)} \quad \textsc{SurfChk_tuple}$$

$$\frac{\begin{array}{l} L : \tau\, \mathbf{in}\, D(\tau_1, .., \tau_i)\, \mathbf{in}\, \Sigma_c \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau\ [\mathit{eff}] \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} L\, se \Leftarrow D(\tau_1, .., \tau_i)\ [\mathit{eff}] \rightsquigarrow L\, ce} \quad \textsc{SurfChk_inject}$$

$$\frac{\begin{array}{l} \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} se \Rightarrow \tau\ [\mathit{eff}''] \rightsquigarrow ce' \\ M = \overline{pat{:}\tau, \cdot \to [\cdot][\cdot]\, se} \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \cdot; M \Leftarrow_{\mathit{eff}''} \tau'\ [\epsilon] \rightsquigarrow ce \\ \mathit{eff} = \mathit{eff}'' \vee \epsilon \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{case}\, se\, \mathbf{of}\, \{pat_1 \to se_1 | .. | pat_n \to se_n\} \Leftarrow \tau'\ [\mathit{eff}] \rightsquigarrow \mathbf{let}\, y = ce'; ce} \quad \textsc{SurfChk_case}$$

$$\frac{\begin{array}{l} \mathbf{1} \leq \mathit{eff}_0 \\ \Sigma_c; \Gamma \vdash_{\mathbf{0}} se_1 \Rightarrow A\ [\mathbf{0}] \rightsquigarrow ce_1 \\ \mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c \\ M = \overline{pat{:}A, \cdot \to [\cdot][\cdot]\, se} \\ \Sigma_c; \Gamma \vdash_{\mathbf{1}} z, \cdot; M \Leftarrow_{\mathbf{0}} D(A, B)\ [\mathit{eff}'] \rightsquigarrow ce_2 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{iter}\, (pat = se_1)\{se_2\} \Leftarrow B\ [\mathbf{1}] \rightsquigarrow \mathbf{iter}\, (z = ce_1)\{ce_2\}} \quad \textsc{SurfChk_iter}$$

$$\frac{\begin{array}{l} \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} se_1 \Leftarrow \mathsf{bool}\ [\mathit{eff}_1] \rightsquigarrow ce_1 \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_2 \Leftarrow \tau\ [\mathit{eff}_2] \rightsquigarrow ce_2 \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_3 \Leftarrow \tau\ [\mathit{eff}_3] \rightsquigarrow ce_3 \\ \mathit{eff} = \mathit{eff}_1 \vee \mathit{eff}_2 \vee \mathit{eff}_3 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{if}\, se_1\, \mathbf{then}\, se_2\, \mathbf{else}\, se_3 \Leftarrow \tau\ [\mathit{eff}] \rightsquigarrow \mathbf{if}\, ce_1\, \mathbf{then}\, ce_2\, \mathbf{else}\, ce_3} \quad \textsc{SurfChk_if}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau\ [\mathit{eff}] \rightsquigarrow ce}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{return}\, se \Leftarrow \mathsf{pred}\ \tau\ [\mathit{eff}] \rightsquigarrow \mathbf{return}\, ce} \quad \textsc{SurfChk_return}$$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_1 \Rightarrow \mathsf{pred}\ \tau\ [\mathit{eff}'] \rightsquigarrow ce_1 \\ \epsilon = \lfloor \mathit{eff}' \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \cdot; pat : \tau \to [\cdot][[\cdot]] se_2 \Leftarrow_\epsilon \mathsf{pred}\ \tau'\ [\mathit{eff}''] \rightsquigarrow ce_2 \\ \mathit{eff} = \mathit{eff}' \vee \mathit{eff}'' \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{take}\, pat = se_1; se_2 \Leftarrow \mathsf{pred}\ \tau'\ [\mathit{eff}] \rightsquigarrow \mathbf{take}\, y = ce_1; ce_2} \quad \textsc{SurfChk_take}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Rightarrow \tau\ [\mathit{eff}] \rightsquigarrow ce}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau\ [\mathit{eff}] \rightsquigarrow ce} \quad \textsc{SurfChk_sub}$$

$\boxed{\begin{array}{l}\Sigma_s \vdash prog \\ \quad \rightsquigarrow \Sigma_c \vdash cprog\end{array}}$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}} se \Leftarrow \tau\ [\mathit{eff}'] \rightsquigarrow ce}{\begin{array}{c}\Sigma_s \vdash \mathsf{main} : \tau_{\mathit{eff}} = se \\ \rightsquigarrow \Sigma_c \vdash \mathsf{main} : \tau_{\mathit{eff}} = ce\end{array}} \quad \textsc{Prog_main}$$

$$\frac{\begin{array}{l} M = \overline{pat{:}A, \cdot \to [\cdot][\cdot]\, se} \\ \Sigma_c; \Gamma \vdash_{\mathbf{0}} y, \cdot; M \Leftarrow_{\mathbf{0}} B\ [\mathit{eff}'] \rightsquigarrow ce \\ \Sigma_s, \mathsf{fun}\ f : A \to_{\mathbf{0}} B = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \vdash prog \\ \quad \rightsquigarrow \Sigma_c, \mathsf{fun}\ f(y : A) \to_{\mathbf{0}} B = ce \vdash cprog \end{array}}{\begin{array}{c}\Sigma_s \vdash \mathsf{fun}\ f : A \to_{\mathbf{0}} B = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \quad prog \\ \rightsquigarrow \Sigma_c \vdash \mathsf{fun}\ f(y : A) \to_{\mathbf{0}} B = ce \quad cprog\end{array}} \quad \textsc{Prog_purefun}$$

$$\frac{\begin{array}{l} M = \overline{pat{:}A, \cdot \to [\cdot][\cdot]\, se} \\ \Sigma_c, f : A \to_{\mathbf{1}} B; \Gamma \vdash_{\mathbf{1}} y, \cdot; M \Leftarrow_{\mathbf{0}} B\ [\mathit{eff}'] \rightsquigarrow ce \\ \Sigma_s, f : A \to_{\mathbf{1}} B \vdash prog \\ \quad \rightsquigarrow \Sigma_c, f : A \to_{\mathbf{1}} B \vdash cprog \end{array}}{\begin{array}{c}\Sigma_s \vdash \mathsf{fun}\ f : A \to_{\mathbf{1}} B = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \quad prog \\ \rightsquigarrow \Sigma_c \vdash \mathsf{fun}\ f(y : A) \to_{\mathbf{1}} B = ce \quad cprog\end{array}} \quad \textsc{Prog_impurefun}$$

$$\frac{\begin{array}{l} M = \overline{pat{:}\tau, \cdot \to [\cdot][\cdot]\, se} \\ \Sigma_c, f : \tau \to_{\mathbf{S}} \tau'; \Gamma \vdash_{\mathbf{S}} y, \cdot; M \Leftarrow_{\mathbf{S}} \tau'\ [\mathit{eff}'] \rightsquigarrow ce \\ \Sigma_s, \mathsf{fun}\ f : \tau \to_{\mathbf{S}} \tau' = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \vdash prog \\ \quad \rightsquigarrow \Sigma_c, \mathsf{fun}\ f(y : \tau) \to_{\mathbf{S}} \tau' = ce \vdash cprog \end{array}}{\begin{array}{c}\Sigma_s \vdash \mathsf{fun}\ f : \tau \to_{\mathbf{S}} \tau' = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \quad prog \\ \rightsquigarrow \Sigma_c \vdash \mathsf{fun}\ f(y : \tau) \to_{\mathbf{S}} \tau' = ce \quad cprog\end{array}} \quad \textsc{Prog_specfun}$$

$$\frac{\begin{array}{l} \Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok} \\ \Sigma_s, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} \vdash prog \\ \quad \rightsquigarrow \Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} \vdash cprog \end{array}}{\begin{array}{c}\Sigma_s \vdash \mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} prog \\ \rightsquigarrow \Sigma_c \vdash \mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} cprog\end{array}} \quad \textsc{Prog_sortdecl}$$

$$\frac{\begin{array}{l} \Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok} \\ \Sigma_s, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} \vdash prog \\ \quad \rightsquigarrow \Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} \vdash cprog \end{array}}{\begin{array}{c}\Sigma_s \vdash \mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 | .. | L_n : A_n\} prog \\ \rightsquigarrow \Sigma_c \vdash \mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 | .. | L_n : A_n\} cprog\end{array}} \quad \textsc{Prog_typedecl}$$

$\boxed{M\ \mathsf{has_con}}$

$$\frac{}{L\, pat : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_con}} \quad \textsc{HC_here}$$

$$\frac{M\ \mathsf{has_con}}{\beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_con}} \quad \textsc{HC_there}$$

$\boxed{M\ \mathsf{has_tup}}$

$$\frac{}{(pat_1, .., pat_n) : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_tup}} \quad \textsc{HT_here}$$

$$\frac{M\ \mathsf{has_tup}}{\beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_tup}} \quad \textsc{HT_there}$$

$\boxed{M/var \overset{y}{\rightsquigarrow}_{eff} M'}$

$$\frac{}{\cdot/var \overset{y}{\rightsquigarrow}_{eff} \cdot} \quad \text{SV_EMPTY}$$

$$\frac{M/var \overset{y}{\rightsquigarrow}_{eff} M'}{x : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M/var \overset{y}{\rightsquigarrow}_{eff} \beta \to [\Gamma, x{:}_{eff}\tau][\mathbf{let}\, x = y; \mathcal{E}]se \mid M'} \quad \text{SV_CONS}$$

$\boxed{M/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} M'}$

$$\frac{}{\cdot/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} \cdot} \quad \text{SC_EMPTY}$$

$$\frac{M/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} M'}{L\, pat : \tau', \beta \to [\Gamma][\mathcal{E}]se \mid M/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} pat : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M'} \quad \text{SC_MATCH}$$

$$\frac{L \neq L' \quad M/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} M'}{L'\, pat : \tau', \beta \to [\Gamma][\mathcal{E}]se \mid M/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} M'} \quad \text{SC_SKIP}$$

$$\frac{M/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} M'}{x : \tau', \beta \to [\Gamma][\mathcal{E}]se \mid M/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} z : \tau, \beta \to [\Gamma, x{:}_{eff}\tau'][\mathbf{let}\, x = y; \mathcal{E}]se \mid M'} \quad \text{SC_VAR}$$

$\boxed{M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{eff} M'}$

$$\frac{}{\cdot/\mathbf{tup} \overset{y}{\rightsquigarrow}_{eff} \cdot} \quad \text{ET_EMPTY}$$

$$\frac{M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{eff} M'}{(pat_1, .., pat_k) : (\tau_1, .., \tau_k), \beta \to [\Gamma][\mathcal{E}]se \mid M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{eff} pat_1 : \tau_1, .., pat_k : \tau_k, \beta \to [\Gamma][\mathcal{E}]se \mid M'} \quad \text{ET_TUPLE}$$

$$\frac{M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{eff} M'}{x : (\tau_1, .., \tau_k), \beta \to [\Gamma][\mathcal{E}]se \mid M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{eff} z_1 : \tau_1, .., z_k : \tau_k, \beta \to [\Gamma, x{:}_{eff}(\tau_1, .., \tau_k)][\mathbf{let}\, x = y; \mathcal{E}]se \mid M'} \quad \text{ET_VA}$$

$\boxed{\Sigma_c; \Gamma \vdash_{eff_0} \overline{y}; M \Leftarrow_\epsilon \tau\ [eff] \rightsquigarrow ce}$

$$\frac{\Sigma_c; \Gamma, \Gamma' \vdash_{eff_0} se \Leftarrow \tau\ [eff] \rightsquigarrow ce' \quad ce = \mathcal{E}[ce']}{\Sigma_c; \Gamma \vdash_{eff_0} \cdot; \cdot \to [\Gamma'][\mathcal{E}]se \mid M \Leftarrow_\epsilon \tau\ [eff] \rightsquigarrow ce} \quad \text{Cov_DONE}$$

$$\frac{\begin{array}{l} M/var \overset{y}{\rightsquigarrow}_\epsilon M' \\ \Sigma_c; \Gamma \vdash_{eff_0} \overline{y}; M' \Leftarrow_\epsilon \tau\ [eff] \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{eff_0} y, \overline{y}; M \Leftarrow_\epsilon \tau\ [eff] \rightsquigarrow ce} \quad \text{Cov_VAR}$$

$$\frac{\begin{array}{l} M\ \mathsf{has_con} \\ D(\overline{a}) = \{\overline{L : \tau}\} \in .. \\ L_1 : \tau_1''\ \mathbf{in}\ D(\tau_1, .., \tau_i)\ \mathbf{in}\ \Sigma_c \quad .. \quad L_n : \tau_n''\ \mathbf{in}\ D(\tau_1, .., \tau_i)\ \mathbf{in}\ \Sigma_c \\ M/L_1{:}\tau_1'' \overset{y}{\rightsquigarrow}_\epsilon M_1 \quad .. \quad M/L_n{:}\tau_n'' \overset{y}{\rightsquigarrow}_\epsilon M_n \\ \Sigma_c; \Gamma \vdash_{eff_0} x_1, \overline{y}; M_1 \Leftarrow_\epsilon \tau\ [eff_1] \rightsquigarrow ce_1 \quad .. \quad \Sigma_c; \Gamma \vdash_{eff_0} x_n, \overline{y}; M_n \Leftarrow_\epsilon \tau\ [eff_n] \rightsquigarrow ce_n \\ eff = eff_1 \vee .. \vee eff_n \end{array}}{\Sigma_c; \Gamma \vdash_{eff_0} y, \overline{y}; M \Leftarrow_\epsilon \tau\ [eff] \rightsquigarrow \mathbf{case}\, y\, \mathbf{of}\, \{L_1\, x_1 \to ce_1 | .. | L_n\, x_n \to ce_n\}} \quad \text{Cov_CON}$$

$$\frac{\begin{array}{l} M\ \mathsf{has_tup} \\ M/\mathbf{tup} \overset{y}{\rightsquigarrow}_\epsilon M' \\ \Sigma_c; \Gamma \vdash_{eff_0} y_1, .., y_n, \overline{y}; M' \Leftarrow_\epsilon \tau\ [eff] \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{eff_0} y, \overline{y}; M \Leftarrow_\epsilon \tau\ [eff] \rightsquigarrow \mathbf{let}\, (y_1, .., y_n) = y; ce} \quad \text{Cov_TUP}$$

$\boxed{\mathcal{E}[ce] \rightsquigarrow ce'}$

$$\frac{}{[\cdot][ce] \rightsquigarrow ce} \quad \textsc{Fill_hole}$$

$$\frac{\mathcal{E}[ce] \rightsquigarrow ce'}{\mathbf{let}\, x = ce''; \mathcal{E}[ce] \rightsquigarrow \mathbf{let}\, x = ce''; ce'} \quad \textsc{Fill_let}$$

$\boxed{\Delta \vdash x \Rightarrow X \dashv \Delta'}$

$$\frac{}{\Delta, x{:}_{\mathit{eff}}\tau \vdash x \Rightarrow \tau \dashv \Delta, x{:}_{\mathit{eff}}\tau} \quad \textsc{RLk_var_here}$$

$$\frac{}{\Delta, \mathsf{log}\ x{:}ce \vdash x \Rightarrow ce \dashv \Delta, \mathsf{log}\ x{:}ce} \quad \textsc{RLk_log_here}$$

$$\frac{}{\Delta, \mathsf{res}\ x{:}^{\mathbf{1}} ce@ce' \vdash x \Rightarrow ce@ce' \dashv \Delta, \mathsf{res}\ x{:}^{\mathbf{0}} ce@ce'} \quad \textsc{RLk_res_here}$$

$$\frac{x \neq y \quad \Delta \vdash x \Rightarrow X \dashv \Delta'}{\Delta, y{:}_{\mathit{eff}}\tau \vdash x \Rightarrow X \dashv \Delta', y{:}_{\mathit{eff}}\tau} \quad \textsc{RLk_skip_var}$$

$$\frac{x \neq y \quad \Delta \vdash x \Rightarrow X \dashv \Delta'}{\Delta, \mathsf{log}\ y{:}ce \vdash x \Rightarrow X \dashv \Delta', \mathsf{log}\ y{:}ce} \quad \textsc{RLk_skip_log}$$

$$\frac{x \neq y \quad \Delta \vdash x \Rightarrow X \dashv \Delta'}{\Delta, \mathsf{res}\ y{:}^{u} ce@ce' \vdash x \Rightarrow X \dashv \Delta', \mathsf{res}\ y{:}^{u} ce@ce'} \quad \textsc{RLk_skip_res}$$

$\boxed{\Sigma_c \vdash \Delta\ \mathsf{wf}}$

$$\frac{}{\Sigma_c \vdash \cdot\ \mathsf{wf}} \quad \textsc{RWf_empty}$$

$$\frac{\Sigma_c \vdash \Delta\ \mathsf{wf}}{\Sigma_c \vdash \Delta, x{:}_{\mathit{eff}}\tau\ \mathsf{wf}} \quad \textsc{RWf_var}$$

$$\frac{\Sigma_c \vdash \Delta\ \mathsf{wf} \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{bool}\ [\mathit{eff}]}{\Sigma_c \vdash \Delta, \mathsf{log}\ x{:}ce\ \mathsf{wf}} \quad \textsc{RWf_log}$$

$$\frac{\Sigma_c \vdash \Delta\ \mathsf{wf} \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce' \Rightarrow \tau\ [\mathit{eff}'] \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{pred}\ \tau\ [\mathit{eff}]}{\Sigma_c \vdash \Delta, \mathsf{res}\ x{:}^{u} ce@ce'\ \mathsf{wf}} \quad \textsc{RWf_res}$$

```
Definition rules:        132 good    0 bad
Definition rule clauses: 301 good    0 bad
```